

SWOMEN TECHSTERS BOOTCAMP 5.1 CAPSTONE PROJECT PROPOSAL

PROJECT TITLE: AI-POWERED SME BUSINESS INTELLIGENCE DASHBOARD; SIMPLIFYING DATA ANALYTICS FOR BETTER BUSINESS DECISIONS

TEAM: GROUP 18

LEARNING TRACKS: DATA ANALYSIS & TECHNICAL PROJECT MANAGEMENT

1. Project Background

Small and Medium Enterprises (SMEs) play a vital role in Nigeria's economy by creating jobs, driving innovation, and contributing significantly to economic growth. Every day, these businesses generate valuable data through sales transactions, customer purchases, inventory records, and financial activities. However, much of this data remains underutilized because many SME owners lack simple tools to transform it into meaningful business insights.

This project aims to develop an AI-Powered SME Business Intelligence Dashboard that helps business owners visualize their sales performance, understand customer behavior, monitor key business metrics, and receive AI-generated insights that support informed decision-making.

2. Problem Statement

Many Small and Medium Enterprises (SMEs) in Nigeria generate valuable business data every day through sales transactions, customer records, and inventory management. However, this data is often stored in spreadsheets or point-of-sale systems without being analyzed to support business decisions.

Most SME owners are not data analysts and often lack the time, technical skills, or tools needed to interpret their business data. As a result, they often rely on experience or intuition when making decisions about sales, inventory, customer management, and business growth.

This project seeks to address this challenge by developing a simple AI-Powered Business Intelligence Dashboard that transforms raw sales data into easy-to-understand visualizations and AI-generated insights. By presenting business information in a clear and accessible format, the dashboard enables SME owners to make more informed decisions without requiring advanced knowledge of data analytics.

3. Project Goal

To develop an interactive AI-Powered Business Intelligence Dashboard that transforms raw sales data into meaningful insights, enabling SMEs to understand their business performance and make informed, data-driven decisions.

4. Project Objectives

The project aims to:

- Clean and organize sales data.
- Analyze business performance using Excel and Power BI.
- Develop an interactive dashboard that presents key business metrics and trends.
- Use AI to generate simple business insights and recommendations.
- Present business information in a clear and easy-to-understand format for SME owners.

5. Target Users

- Small and Medium Enterprise (SME) owners
- Retail business owners
- Entrepreneurs
- Business managers
- Startups

6. Proposed Solution

The team will develop an interactive Business Intelligence Dashboard using Microsoft Excel and Power BI with a sample SME sales dataset.

The dashboard will provide business owners with a simple way to monitor business performance through easy-to-understand visualizations and key performance indicators.

Executive Dashboard

- Total Sales
- Total Profit

- Number of Orders
- Monthly Sales Trend

Product Analysis

- Best-selling products
- Most profitable products
- Sales by product category

Customer Analysis

- Top customers
- Customer segments
- Regional sales performance

The dashboard will transform raw sales data into meaningful business insights, enabling users to quickly identify trends, monitor performance, and make informed business decisions.

7. AI Component

Artificial Intelligence will enhance the solution by making business insights easier to understand.

ChatGPT and Claude AI will be used to:

- Generate summaries of dashboard findings.
- Explain sales trends in simple language.
- Suggest practical business improvements based on the analyzed data.
- Answer business questions such as:
 - Which products generate the highest sales?
 - Which region performs best?
 - Which products need improvement?
 - What actions can improve business performance?

8. Sustainable Development Goals (SDGs)

This project aligns with:

SDG 8 – Decent Work and Economic Growth

Supporting SMEs with data-driven decision-making to improve productivity, innovation, and business growth.

SDG 9 – Industry, Innovation and Infrastructure

Promoting digital transformation through Business Intelligence and Artificial Intelligence.

9. Technology Stack

- Microsoft Excel (Data Cleaning and Preparation)
- Microsoft Power BI (Dashboard Development and Data Visualization)
- ChatGPT (AI-generated Business Insights)
- GitHub (Project Repository)
- Microsoft Teams (Team Collaboration)
- Kaggle

10. Expected Deliverables

The team will deliver:

- Cleaned SME sales dataset
- Interactive Power BI Dashboard
- AI-generated business insights and recommendations
- Project documentation
- GitHub repository
- Demo Day presentation slides

11. Team Roles and Responsibilities

Role	Responsibility	Team Member Name
Project Lead	Coordinate project activities, assign tasks, monitor progress, and ensure timely delivery.	Ayomide Elizabeth Temidayo
Technical Project Manager	Coordinate project activities, assign tasks, Develop project schedule, monitor milestones, manage risks, and facilitate team collaboration and ensure timely delivery..	Morenikeji Bankole
Data Analyst(s)	Collect, clean, organize, and analyze sales data.	1. Chiedza Rene Mashaire - Unit Head 2. Olashile Ogundimu 3. Cynthia Amaka 4. Mwape Lugali Chishimba 5. Ayomide Elizabeth Temidayo
Dashboard Developer	Design and develop the interactive Power BI dashboard.	1. Elizabeth Danladi Sabatu - U/Head
AI Support	Generate business insights and recommendations using ChatGPT and Claud AI	1. Nmesoma Igbokwe - U/Head 2. Falayi Omotolani Olajumoke 3. Chioma Loveth Nwankwo
Documentation Lead	Prepare project proposals, documentation, and final report.	1. Oladiran Racheal Bolanle - Unit Head 2. Morenikeji Bankole 3. Sophie Damilola Daniels 4. Olabode Olasumbo
Presentation Lead	Develop presentation slides and coordinate Demo Day presentations.	Ayomide Elizabeth Temidayo Morenikeji Bankole
Quality Assurance	Review dashboard accuracy, validate findings, and ensure quality before submission.	1. Jadesola Fawibe - Unit Head Washika Comfort Msurshima

12. Project Timeline

Activity	Timeline
Project Planning	Day 1
Data Collection & Cleaning	Day 1–2
Data Analysis	Day 2–3
Dashboard Development	Day 3–5
AI Integration & Documentation	Day 5–6
Testing & Final Review	Day 6–7
Final Submission	Day 7
Demo Day Presentation	Day 8

13. Expected Impact

- The AI-Powered SME Business Intelligence Dashboard will help SME owners better understand their business by converting raw sales data into clear visual insights and practical recommendations.
- Democratizing data analytics for SMEs by providing a simple and user-friendly solution that enables business owners regardless of their technical background to understand their business data and make informed decisions.
- Lay the foundation for a future self-service business intelligence platform that empowers SME owners to independently analyze and understand their business data using AI-assisted analytics.

The project will enable business owners to:

- Monitor sales and business performance more effectively.
- Understand customer and product trends.
- Make informed business decisions using data rather than guesswork.
- Identify opportunities for business growth.
- Improve operational efficiency through simple, data-driven insights.

Ultimately, the project demonstrates how data analytics and Artificial Intelligence can simplify decision-making for SMEs, making business intelligence more accessible to entrepreneurs who may not have technical expertise.

NAMES OF ACTIVE PARTICIPANT ON THE PROJECT

S/N	NAME	STUDENT ID	EMAIL
1	Ayomide Elizabeth Temidayo		
2	Morenikeji Bankole	WTB/2026/5.1/0019	
3	Elizabeth Danladi Sabatu		
4	Chiedza Rene Mashaire		
5	Olashile Ogundimu		
6	Peter Ukamaka Cynthia		
7	Falayi Omotolani Olajumoke		
8	Nmesoma Igbokwe		
9	Chioma Loveth Nwankwo		
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